```
CREATE DATABASE IF NOT EXISTS ass4;
USE ass5;

-- Step 1 : Drop existing tables for re-run
DROP TABLE IF EXISTS Fine;
DROP TABLE IF EXISTS Borrower;

-- Step 2 : Create Borrower table
CREATE TABLE Borrower (
   Rollno INT PRIMARY KEY,
   Name VARCHAR(100),
   DateofIssue DATE,
   NameofBook VARCHAR(200),
   Status CHAR(1)   -- 'I' = Issued, 'R' = Returned
);

-- Step 3 : Create Fine table (avoid reserved word 'date')
CREATE TABLE Fine (
   Roll_no INT,
   FineDate DATE,
   Amt DECIMAL(10,2),
   FOREIGN KEY (Roll_no) REFERENCES Borrower(Rollno)
);

-- Step 4 : Insert sample data
INSERT INTO Borrower (Rollno, Name, DateofIssue, NameofBook, Status) VALUES
(101, 'Aman',  '2025-07-01', 'Database Systems', 'R'),
(102, 'Priya', '2025-06-10', 'Operating Systems', 'I'),
(103, 'Ravi',  '2025-07-10', 'Computer Networks', 'I'),
(104, 'Neha',  '2025-06-01', 'Java Programming', 'I'),
(105, 'Arjun', '2025-07-01', 'Data Structures', 'I'),
(106, 'Sneha', '2025-07-20', 'Machine Learning', 'I'),
(107, 'Rahul', '2025-06-15', 'Cloud Computing', 'R');

INSERT INTO Fine (Roll_no, FineDate, Amt)
VALUES (107, '2025-07-10', 325.00);

-- Step 5 : Create Stored Procedure
DELIMITER //

CREATE PROCEDURE return_book (
  IN p_rollno INT,
  IN p_bookname VARCHAR(200)
)
BEGIN
   DECLARE v_date_of_issue DATE;
   DECLARE v_days INT DEFAULT 0;
   DECLARE v_fine DECIMAL(10,2) DEFAULT 0.00;
   DECLARE v_status CHAR(1);

   DECLARE EXIT HANDLER FOR SQLEXCEPTION
   BEGIN
      SELECT 'An error occurred. Please check inputs or try again.' AS ErrorMessage;
   END;
```

```
    -- Try fetching the issue info
    SELECT DateofIssue, Status
    INTO v_date_of_issue, v_status
    FROM Borrower
    WHERE Rollno = p_rollno AND NameofBook = p_bookname
    LIMIT 1;

    -- If no matching record found
    IF v_date_of_issue IS NULL THEN
        SELECT CONCAT('No issued record found for Rollno=', p_rollno, ' and Book=', p_bookname) AS
Message;
    ELSE
        -- If already returned
        IF v_status = 'R' THEN
            SELECT 'Book has already been returned.' AS Message;
        ELSE
            -- Calculate fine
            SET v_days = DATEDIFF(CURDATE(), v_date_of_issue);

            IF v_days > 30 THEN
                SET v_fine = (15 * 5) + ((v_days - 30) * 50);
            ELSEIF v_days > 15 THEN
                SET v_fine = (v_days - 15) * 5;
            ELSE
                SET v_fine = 0;
            END IF;

            -- Update borrower status
            UPDATE Borrower
            SET Status = 'R'
            WHERE Rollno = p_rollno AND NameofBook = p_bookname;

            -- Insert fine if applicable
            IF v_fine > 0 THEN
                INSERT INTO Fine (Roll_no, FineDate, Amt)
                VALUES (p_rollno, CURDATE(), v_fine);
            END IF;

            -- Final message
            SELECT 'Book returned successfully.' AS Message, v_days AS Days_Borrowed, v_fine AS Fine;
        END IF;
    END IF;
END;
//

DELIMITER ;

-- Step 6 : Test Procedure
CALL return_book(102, 'Operating Systems');
CALL return_book(107, 'Cloud Computing');

-- Step 7 : View Tables
SELECT * FROM Borrower;
SELECT * FROM Fine;
```

-- ----------
-- 💬 Viva Preparation Notes
-- ----------
-- 🔹 1Purpose:
-- To automate fine calculation and update borrower status
-- whenever a book is returned.

-- 🔹 2What is DATEDIFF() used for?
-- ➤ It calculates the number of days between the issue date and current date.

-- 🔹 3Why use EXIT HANDLER?
-- ➤ To catch SQL errors and prevent the procedure from stopping abruptly.

-- 🔹 4Why use IF v_date_of_issue IS NULL?
-- ➤ To check if the given Rollno and Bookname exist in the Borrower table.

-- 🔹 5Explain the fine logic:
-- ➤ No fine for 15 days, ₹5/day for next 15 days, and ₹50/day after 30 days.

-- 🔹 6Why use CURDATE()?
-- ➤ CURDATE() returns the system's current date, used to calculate delay.

-- 🔹 7Why Status = 'I' or 'R'?
-- ➤ 'I' = book is currently issued, 'R' = book has been returned.

-- 🔹 8How can this be automated further?
-- ➤ By adding a TRIGGER to auto-calculate fine when a borrower's status changes to 'R'.

-- 🔹 9Real-Life Application:
-- ➤ Used in library management systems to handle book returns and fine tracking.

-- 🔹 10what will happen if you call return_book twice for the same entry?
-- ➤ The second call will display "Book has already been returned."
-- ----------